Worksheet 8-2 A Particle Motion #1

AP Calculus Name_____

You must show all work. If the problem is calculator active, you must show your setup.

1.) A particle moves along the x-axis for $t \geq 0$ with an acceleration of $a(t) = 12t + 6$ where t is time in seconds. The particle's velocity at $t = 3$ is 36 cm/sec. The initial position of the particle is 4 cm. What is the position of the particle when the velocity is zero?

2.) A particle's velocity along the x-axis is given by $v(t) = 5\cos t$.

(a) Find the particle's displacement on the interval $0 \leq t \leq \frac{3\pi}{2}$.

(b) If $s(0) = 3$, what is the particle's position at $t = \frac{3\pi}{2}$

3.) A particle moves along the y-axis for $t \geq 0$ with a velocity of $v(t) = 12t^2 - 24t$. The particle's initial position is 10 cm. Find the position of the function at the particle's minimum velocity

4.) A particle moves along the y-axis with an acceleration of $a(t) = 12t - 6$ with initial velocity of –10 and the position at $t = 1$ is –9. Find the position function for the particle.

5.) **Calculator Active.** A particle moves along the x-axis. The velocity of the particle at time t is given by $v(t) = \frac{2}{t^2+3}$. If the position of the particle is $x = 2$ when $t = 4$, what is the position of the particle when $t = 6$?

6.) **Calculator Active.** A particle moves along the y-axis with initial position $y(0) = 1$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \cos(\pi t)$. What is the position of the object at time $t = 3$?

7.) **Calculator Active.** A particle moves along the x-axis with a velocity of $v(t) = \sqrt[3]{t^2} - \frac{1}{t^2}$ measured in inches/second. At $t = 1$ the position of the particle is 3 inches. What is the particle's position at $t = 8$?